// Passing cars:

```
FUNCTION PassingCars(A)
    eastbound = 0
    pairs = 0

    FOR EACH car IN A
        IF car = 0 THEN
            eastbound = eastbound + 1
        Else
            pairs = pairs + eastbound
        END IF

        IF pairs > 1000000000 THEN
            RETURN -1
        END IF
    END FOR

    RETURN pairs
END FUNCTION
```

2/ PermCheck :

```
FUNCTION PermCheck(A)
    N = LENGTH(A)
    seen = EMPTY SET

    FOR EACH number IN A
        IF number < 1 OR number > N THEN
            RETURN 0
        END IF

        IF number IS IN seen THEN
            RETURN 0
        END IF

        ADD number TO seen
    END FOR

    RETURN 1
END FUNCTION
```

8/ Tape Equilibrium:

```
FUNCTION TapeEquilibrium(A)
    total = SUM(A)
    left = 0
    best = INFINITY

    FOR i FROM 0 TO LENGTH(A) - 2
        left = left + A[i]
        right = total - left

        difference = ABS(left - right)

        IF difference < best THEN
            best = difference
        END IF
    END FOR

    RETURN best
END FUNCTION
```

4/ FrogJmp:

```
FUNCTION FrogJmp(X, Y, D)
    distance = Y - X

    jumps = distance DIV D

    IF distance MOD D IS NOT 0 THEN
        jumps = jumps + 1
    END IF

    RETURN jumps
END FUNCTION
```

5/ OddOccurrencesInArray:

```
FUNCTION OddOccurrencesInArray(A)
    result = 0

    FOR EACH number IN A
        result = result XOR number
    END FOR

    RETURN result
END FUNCTION
```